FULL CARD 25 — Forces, Mass and Acceleration

PHYSICS • NEWTON'S SECOND LAW • FREE-BODY REASONING

MODEL • Resultant force determines acceleration: F = ma. If the resultant force is zero, velocity remains constant (including the possibility of remaining at rest).

Force investigation: trolley motion

Trial	Mass (kg)	Resultant force (N)	Acceleration (m/s²)	F ÷ m
A	0.50	0.50	1.0	1.0
B	0.50	1.00	2.0	2.0
C	1.00	1.00	1.0	1.0
D	1.50	1.00	0.67	0.67

Worked calculations

Trial B: a = F ÷ m = 1.00 N ÷ 0.50 kg = 2.0 m/s². Trial D: a = 1.00 ÷ 1.50 ≈ 0.67 m/s². For a fixed mass, doubling resultant force doubles acceleration; for fixed force, increasing mass reduces acceleration.

Free-body reasoning

Situation	Forces to consider	Reasoning
Book resting on table	Weight downward; normal contact force upward	Forces balance vertically
Falling object with air resistance	Weight downward; drag upward	Resultant depends on relative sizes
Car braking	Braking force opposite motion; other resistive forces	Speed decreases if resultant opposes velocity
Satellite in orbit	Gravity directed toward central body	Gravity changes direction of velocity

Units and conversions

1 N = 1 kg·m/s². Speed: m/s. Acceleration: m/s². Convert grams to kilograms before using F = ma: 500 g = 0.500 kg.

Evaluation: Friction, timing delay, uneven surfaces, and inaccurate mass measurements can affect results. Repeat trials, use a motion sensor if available, and keep other variables controlled.

Practice: A 2 kg cart experiences a 6 N forward force and 2 N friction. Find resultant force and acceleration. State the direction of acceleration.